

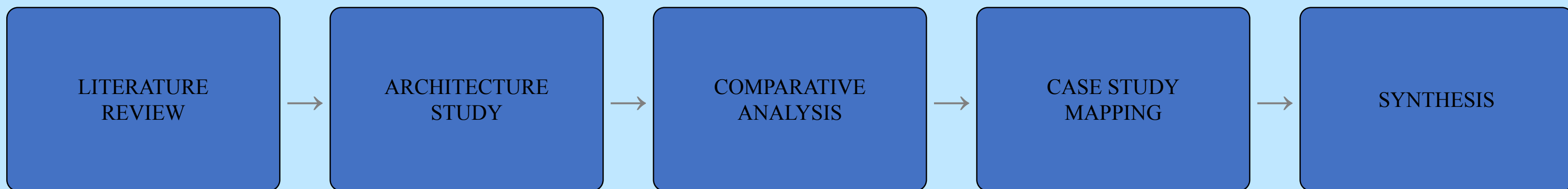


Agentic LLMs: From Chatbots to Autonomous AI Agents

INTRODUCTION

- Traditional chatbots respond passively to single prompts and have no memory, planning ability, or means to act in the real world.
- Agentic LLMs extend a language model with reasoning, planning, memory, and the ability to invoke external tools toward a goal.
- Core building blocks: an LLM reasoning core, a memory/context store, a plan-act-observe loop (e.g. ReAct), and a tool/API interface.
- Applications include coding assistants, autonomous research agents, workflow automation, and multi-agent collaboration.

MATERIALS AND METHODS



Literature-based comparative methodology: agent frameworks and papers were surveyed, their architectures analyzed, and chatbot vs. agentic-LLM capabilities compared across common task dimensions — tool use, reasoning depth, autonomy, and context memory.

RESULTS

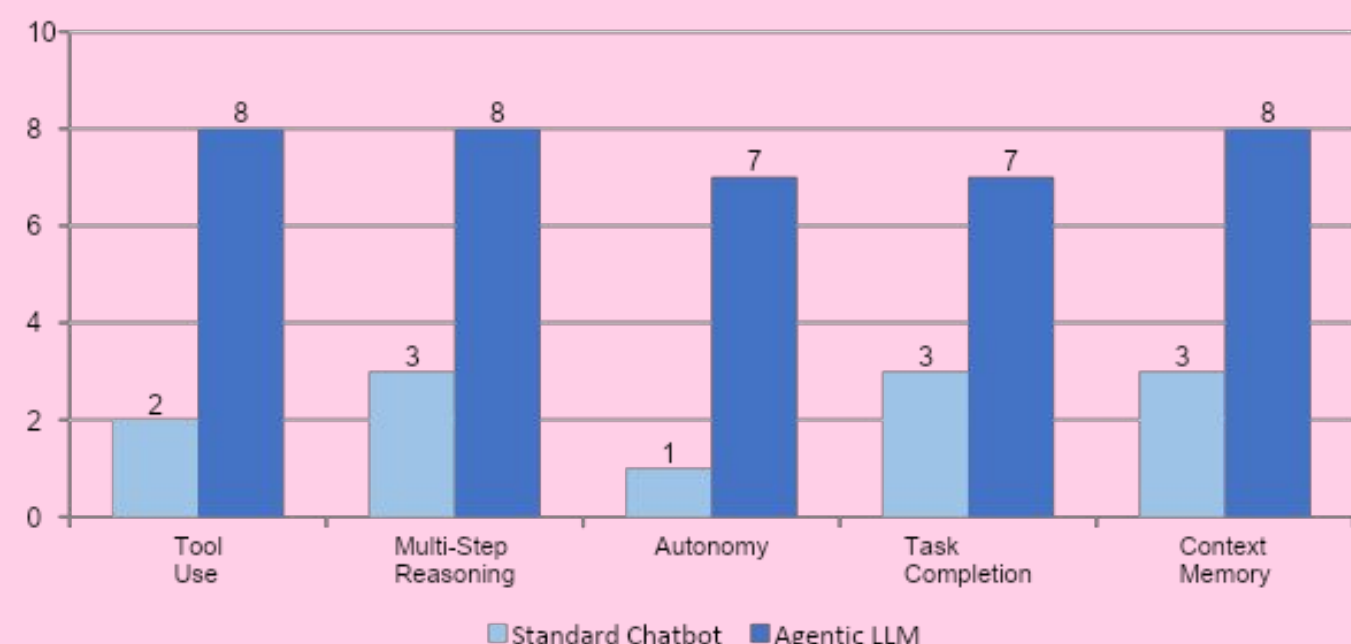


Fig-1: Standard Chatbot vs Agentic LLM Capability Comparison (illustrative)

- Agent-style reasoning (plan → act → observe) consistently outperforms single-shot prompting on multi-step tasks.
- Tool-augmented agents show the largest gains on tasks needing lookup, computation, or verification.
- Autonomy gains taper as task horizon grows — longer plans accumulate compounding errors.
- Adding a critic/verifier step measurably improves final-answer reliability over a single agent loop.

DISCUSSION AND CONCLUSION

- Agentic LLMs mark a shift from reactive text generation to proactive, multi-step task execution.
- Key open challenges: reliable long-horizon planning, compounding errors, and safe, verifiable tool use.
- Planner–executor–critic multi-agent designs improve robustness but add coordination and compute cost.
- Future work should prioritize standardized agent benchmarks and stronger guardrails on autonomous action.

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